Customer Behaviour Analysis

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A deep dive into retail and e-commerce customer data — exploring purchasing patterns, discount effectiveness, product performance, and customer segmentation to drive smarter business decisions.

Dataset Type

Retail / E-commerce — Structured Tabular Data at the Customer Transaction Level

Data Mix

Numerical (Age, Purchase Amount, Previous Purchases) + Categorical (Category, Payment Method, Season)

Core Purpose

Customer behavior analysis and business decision-making across 5,000 customers and 30 unique items

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Domain Background & Business Problem

Why This Data Matters

In the e-commerce industry, companies track every customer interaction — what customers buy, how often they buy, how much they spend, and whether they respond to discounts. This data enables businesses to:

- Identify **high-value customers**
- Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**

The Business Problem

How can a retail company understand customer purchasing behavior to **increase revenue, improve customer retention, and optimize marketing strategies?**

Key Problem Areas

- **Low Customer Retention** — Who are frequent vs. one-time buyers?
- **Ineffective Marketing** — Do discounts and promo codes actually work?
- **Product Demand** — Which categories and products generate the most revenue?
- **Payment Behavior** — Which payment methods are preferred? Are digital payments rising?
- **Seasonal Variation** — Which season drives maximum purchases?
- **Customer Segmentation** — Identify high-value, frequent, and discount-sensitive customers

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Data Dictionary

The dataset contains 17 columns capturing a comprehensive view of each customer transaction, from demographics to payment behavior.

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male / Female / Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer (USD)
Location	Customer's geographic location
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes / No)
Discount Applied	Whether a discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment used
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

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Exploratory Data Analysis (EDA) in Python

Data Loading

Initial Exploration

Consistency Check

Missing Value Handling

The EDA process began by importing `pandas` and loading the dataset with `df = pd.read_csv('customer_shopping_behavior.csv')`. Initial exploration used `df.head()`, `df.tail()`, `df.describe(include='all')`, and `df.info()` to understand data structure, types, and quality issues early. A category correction mapping was then created to fix inconsistencies between **Item Purchased** and **Category** — a critical real-world data cleaning step ensuring accurate grouping and correct business insights.

Size Column

Electronics & Accessories → "Not Applicable"; Footwear → "Free Size"; Clothing → filled with **mode** (domain logic)

Review Rating

Filled using **product-level mean** (not global average) — an advanced imputation technique keeping ratings realistic

Purchase Amount

Replaced with **mean value** to maintain product-level pricing consistency and avoid revenue distortion

Previous Purchases

Replaced with **0** — missing implies no prior purchase; critical for segmentation and loyalty analysis

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Data Cleaning, Standardization & Export

Duplicate Handling

Duplicates were identified using **Customer ID** and removed to ensure each customer appears only once — critical for accurate reporting and avoiding double-counting in analysis.

Column Standardization

Column names were cleaned using:

```
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.lower()
```

This makes column names clean, SQL-friendly, and easy to use in Power BI.

Export to SQL Server

After cleaning, the data was exported to SQL Server for further analysis using:

```
pip install pyodbc sqlalchemy
```

Moving cleaned data into SQL enables:

- Query-based insights at scale
- Integration with BI tools like Power BI
- Structured, repeatable analysis workflows

All cleaning steps reflect real-world domain understanding — not just technical fixes, but business-logic-driven decisions.

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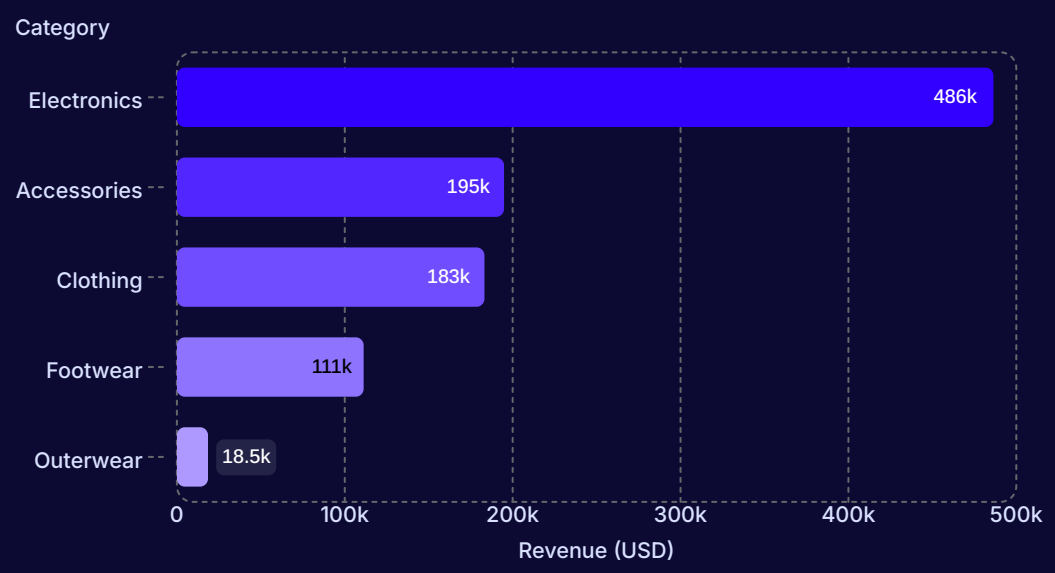
SQL Analysis: Revenue, Discounts & Gender

Revenue by Category



Electronics dominates with $486K — nearly 2.5× the next category (Accessories at $194K).

Do Discounts Increase Purchase Value?

Customers with discounts spend **$219.22 on average vs. $182.24** without — discounts do drive higher spend per transaction.

Revenue by Gender

$391K

Male

$352K

Female

$251K

Other

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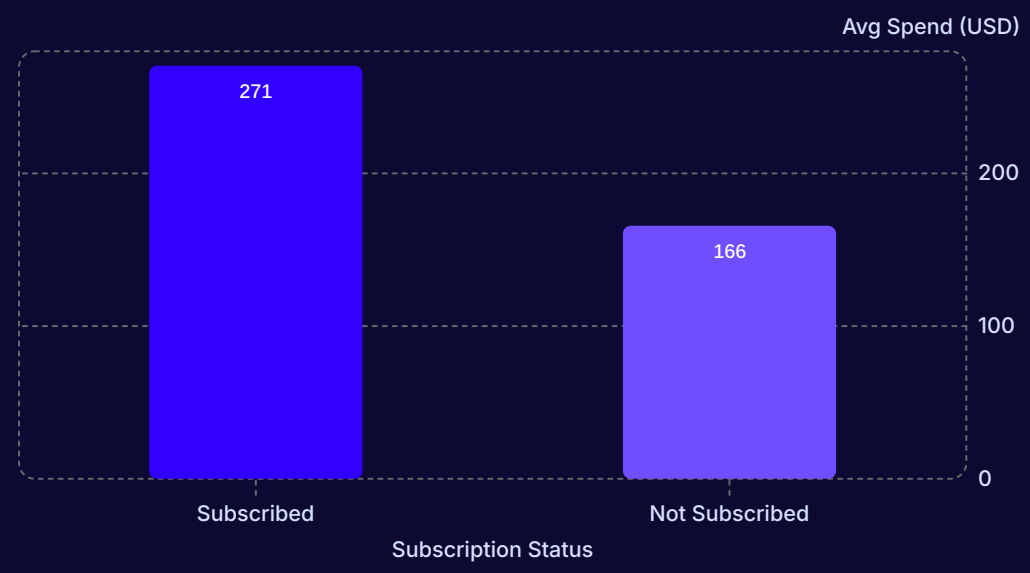
SQL Analysis: Subscriptions, Ratings & Segmentation

Do Subscribed Customers Spend More?



Subscribed customers (1,579) average **$270.58** vs. $165.72 for non-subscribers (3,421) — a **63% higher average spend**. Total revenue: $427K vs. $567K respectively.

Customer Segmentation

3,085

Loyal Customers

1,354

Returning

561

New Customers

Top 5 Products by Avg Review Rating

Gloves 3.86/5	Sandals 3.84/5
Boots 3.82/5	Hat 3.80/5
Skirt 3.79/5	

Bottom 5 Products by Avg Review Rating

Item	Avg Rating
Phone	2.94
Headphones	3.08
Watch	3.10
Bag	3.17
Laptop	3.19

⚠ Electronics dominate both top revenue AND bottom ratings — a key tension for product strategy.

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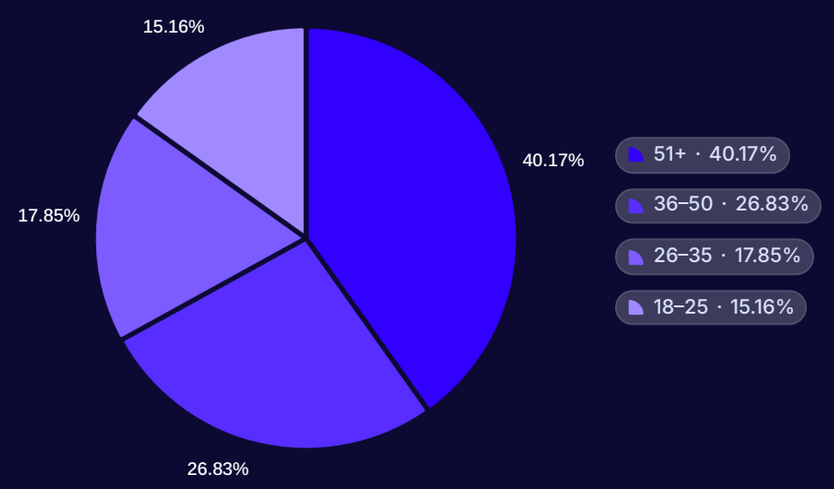
SQL Analysis: Top Products, Age Groups & Repeat Buyers

Revenue by Age Group



Customers aged **51+** contribute the largest revenue share at **40.2% ($399K)**, making older demographics the most valuable segment.

Top Products per Category

Category	Top Item	Units Sold
Clothing	Shirt	315
Footwear	Shoes	303
Electronics	Phone	171
Accessories	Jewelry	171
Outerwear	Jacket	163

Repeat Buyers & Subscription

Of **repeat buyers** (more than 5 previous purchases), **1,190 are subscribed** and 2,763 are not — suggesting a significant opportunity to convert loyal repeat buyers into subscribers. Among normal buyers, 389 subscribe vs. 658 who don't.

Top Discount Usage Products

Laptop (51.7%), Phone (51.5%), and Headphones (51.2%) have the highest discount usage rates — all electronics, reinforcing the category's price-sensitivity.

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Power BI Dashboard

5,000
Total Customers

30
Unique Items

3.62
Avg Rating

$994K
Total Spend

$198.84
Avg Spend

Revenue by Season

Season	Revenue
Spring	$243.88K
Winter	$241.39K
Summer	$236.66K
Autumn	$212.94K

Payment Method Split

Method	Share
Debit Card	25.5%
Cash	23.5%
Credit Card	22.7%
PayPal	20.9%
Venmo / Bank Transfer	7.5%

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Key Findings & Business Recommendations

Electronics = Top Revenue Driver

Electronics generates **$486K** — nearly 50% of total revenue — but also scores the **lowest review ratings**. Improving product quality or post-purchase support is critical.

Discounts Work — Strategically

Discounted customers spend **$219 vs. $182** without discounts. Target discount campaigns at electronics buyers (51%+ usage rate) to maximize ROI.

Subscriptions Drive Higher Spend

Subscribers spend **63% more** on average. With 2,763 repeat buyers not yet subscribed, a targeted subscription conversion campaign could significantly boost CLV.

Focus on 51+ Demographic

Customers aged 51+ contribute **40.2% of total revenue** ($399K). Marketing and product strategies should prioritize this high-value, often underserved segment.

Top revenue locations: New York ($143.9K), Los Angeles ($125.7K), Phoenix ($119.1K). Lowest: Kansas ($3.44K), Hawaii ($3.75K). Geographic targeting can optimize ad spend significantly.

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